Multiple comparisons

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Ask which comparisons belong to the family before inspecting attractive p-values.

Teach the union-bound guarantee without confusing it with false-discovery proportions or useful effects.

Keep one habit: Model five-test decisions → explain union bound → trace six/ten-test scenarios → check missing outcomes → assess four-test transfer.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of **multiple comparisons**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Statistical tests and assumptions. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of statistical tests and assumptions

Reference example, not the only acceptable wording: Teacher explanation: A statistical test compares observed **evidence** with a specified null model. A p-value is not the probability that the null hypothesis is true.

Toy paired sign-flip test assumes independent paired differences whose signs are exchangeable under a symmetric zero-effect null. Enumerate all2^n sign assignments, holding magnitudes fixed.

Two-sided p is fraction with absolute mean at least the observed absolute mean; use inclusive ties.

Without opening the old lesson, explain one key idea from Uncertainty intervals. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of uncertainty intervals

Reference example, not the only acceptable wording: Teacher explanation: An uncertainty interval needs a sampling model and assumptions.

A confidence level refers to the repeated-sampling method, not a probability attached to a fixed unknown parameter after observing data.

For this constructed exercise assume independent normal measurements with known population standard deviation 2. A 95% mean interval uses supplied normal critical value 1.96:mean plus/minus1.96*sigma/sqrt(n).

Unknown sigma requires a different justified method.

Without opening the old lesson, explain one key idea from Experimental units and independence. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of experimental units and independence

Reference example, not the only acceptable wording: Teacher explanation: The experimental unit is the independently assigned or sampled entity. Multiple measurements from one unit are often related.

If methods are compared within each query, queries are paired units. Repeated runs within a query estimate run variation but do not increase the number of independent query contexts.

Pictured end goal

Explain **multiple comparisons** using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Statistical tests and assumptions

Uncertainty intervals

Experimental units and independence

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. Testing many hypotheses creates more chances for false discoveries. A family rule must be specified before selecting attractive p-values.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of **multiple comparisons**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach familywise error with complete denominators and honest missing outcomes

> Distinguish a probability bound, observed decisions and practical effects.

Definitions and family contract

Testing many hypotheses creates more chances for false discoveries. A family rule must be specified before selecting attractive p-values.

For this exercise use Bonferroni: family alpha .05 divided by number of tests m; reject each test when its valid p-value is at most .05/m. This controls the **family error** by a bound without requiring independent tests.

Declared item	Complete exercise contract
Family	All five stated tests; do not discard unfavorable results
Supplied p-values	0.009, 0.012, 0.3, 0.4, 0.8; stipulated valid tests, not fitted from invented data
Family alpha	0.05, fixed before inspecting results
Per-test rule	Reject when $p \le 0.05 / m$, including equality
What is controlled	Probability of at least one false rejection among true nulls is at most 0.05

Declared item	Complete exercise contract
Dependence	Bonferroni union bound does not require independent tests; individual p-values must still be valid
Not established	Effect magnitude, practical value, null truth and actual study quality

Main decisions, union-bound reasoning and teacher key

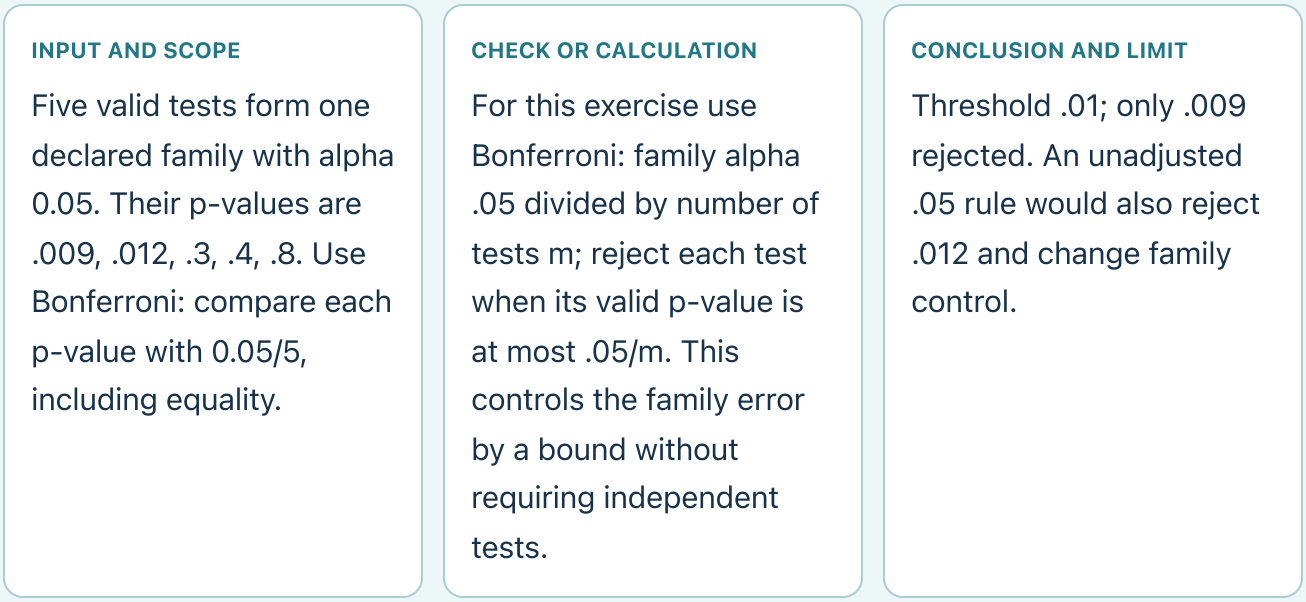
Main test	Supplied p	Bonferroni cutoff	Decision
H1	0.009	0.010	Reject
H2	0.012	0.010	Do not reject
H3	0.300	0.010	Do not reject
H4	0.400	0.010	Do not reject
H5	0.800	0.010	Do not reject
Unadjusted comparison	H1 and H2 are ≤ 0.05	This is a different decision rule	Does not provide the same familywise bound

Why the bound works: with m tests, each true null has false-rejection probability at most alpha/m when its p-value is valid. The probability of any false rejection is at most the sum of those probabilities. There are at most m true nulls, so the sum is at most m × alpha/m = alpha. This is an upper bound, not a measured error frequency, and it does not require independence across tests.

Main: 0.05/5 = 0.01, so only 0.009 rejects; 0.012 would reject only under the unadjusted 0.05 rule. Picture: 0.05/6 = 1/120, so only 0.004 rejects. Changed ten-test family: cutoff 0.005; only 0.004 among the six supplied values rejects. Four missing outcomes cannot be classified. Independent task: 0.05/4 = 0.0125, so only 0.01 rejects. Equality is included.

The union bound needs valid individual p-values, not independent tests; it bounds the chance of any false rejection, not the percentage of reported discoveries that are false. No effect size, actual data or null-truth probability is supplied.

Multiple comparisons — evidence trace



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Threshold .01; only .009 rejected. An unadjusted .05 rule would also reject .012 and change family control. Qualification: Adjustment cannot repair invalid tests, post-hoc outcome selection or unreported analyses.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Five valid tests form one declared family with alpha 0.05. Their p-values are .009, .012, .3, .4, .8. Use Bonferroni: compare each p-value with 0.05/5, including equality.

Threshold .01; only .009 rejected. An unadjusted .05 rule would also reject .012 and change family control.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Family and validity probes

Audit question	Checked answer
p equals alpha/m exactly	Reject under the declared inclusive rule
Keep only the smallest p and call m = 1	Invalid reconstruction of the original family
Tests are dependent	Bonferroni can still control familywise error if each null p-value is valid
Individual p-values are invalid	Dividing alpha cannot repair that defect
Four family results are missing	Keep them in the family size and mark outcomes unknown
A null is rejected	Does not establish a large useful effect or the probability that the null is false
A null is not rejected	Does not prove no effect
Bonferroni is conservative	The bound can be loose; the actual error probability need not equal alpha

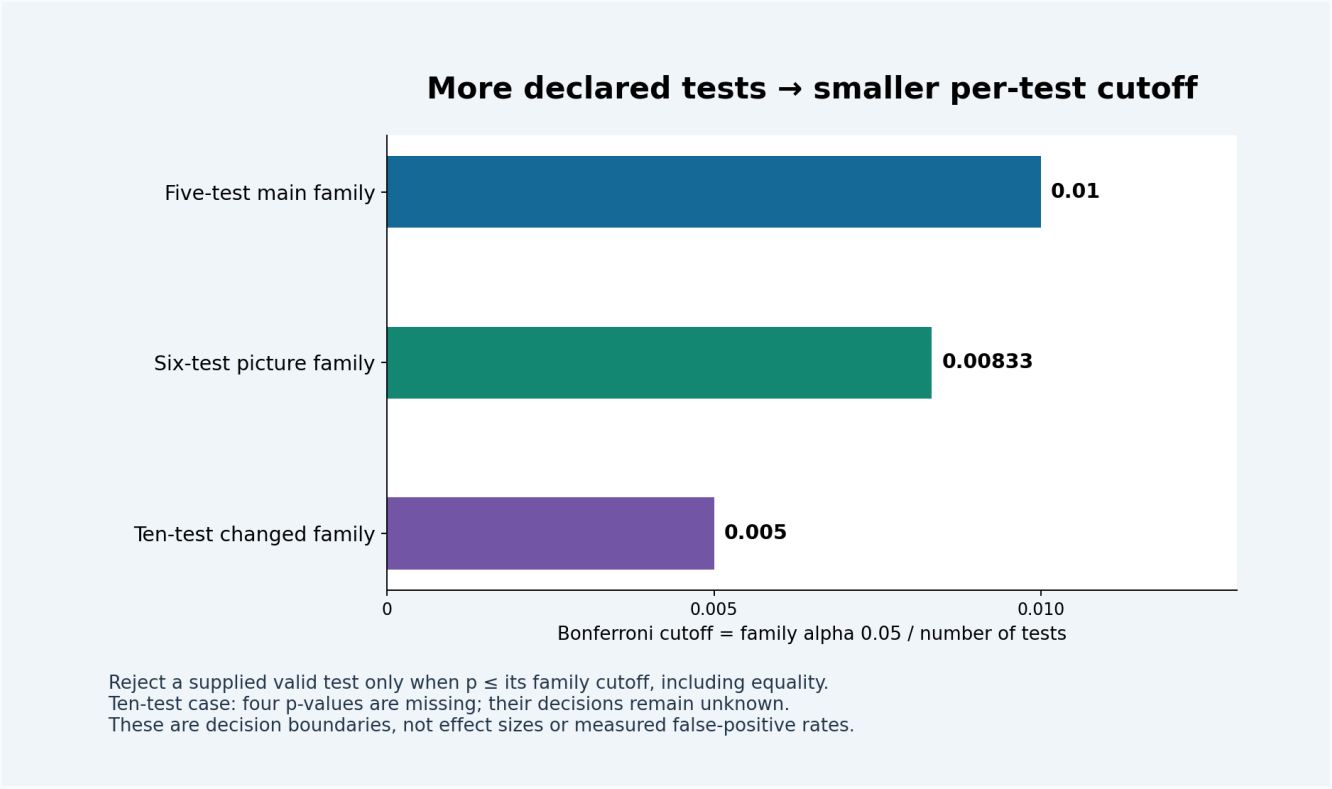
Separate six-test and changed ten-test picture

Six declared prompt comparisons. A team performs six valid hypothesis tests and applies a predeclared Bonferroni familywise threshold.

Familywise alpha=0.05. Supplied unadjusted p-values are 0.004,0.012,0.02,0.08,0.4,0.9. These values are stipulated, not calculated from invented data.

The separate picture has six supplied p-values: 0.004, 0.012, 0.02, 0.08, 0.4 and 0.9. Its exact cutoff is 0.05/6 = 1/120, approximately 0.00833. Only 0.004 rejects. In the changed scenario, ten tests were declared but only these six values are supplied. Use all ten in the denominator: the cutoff is 0.005. Only 0.004 among the supplied values rejects; the other four decisions remain unknown.

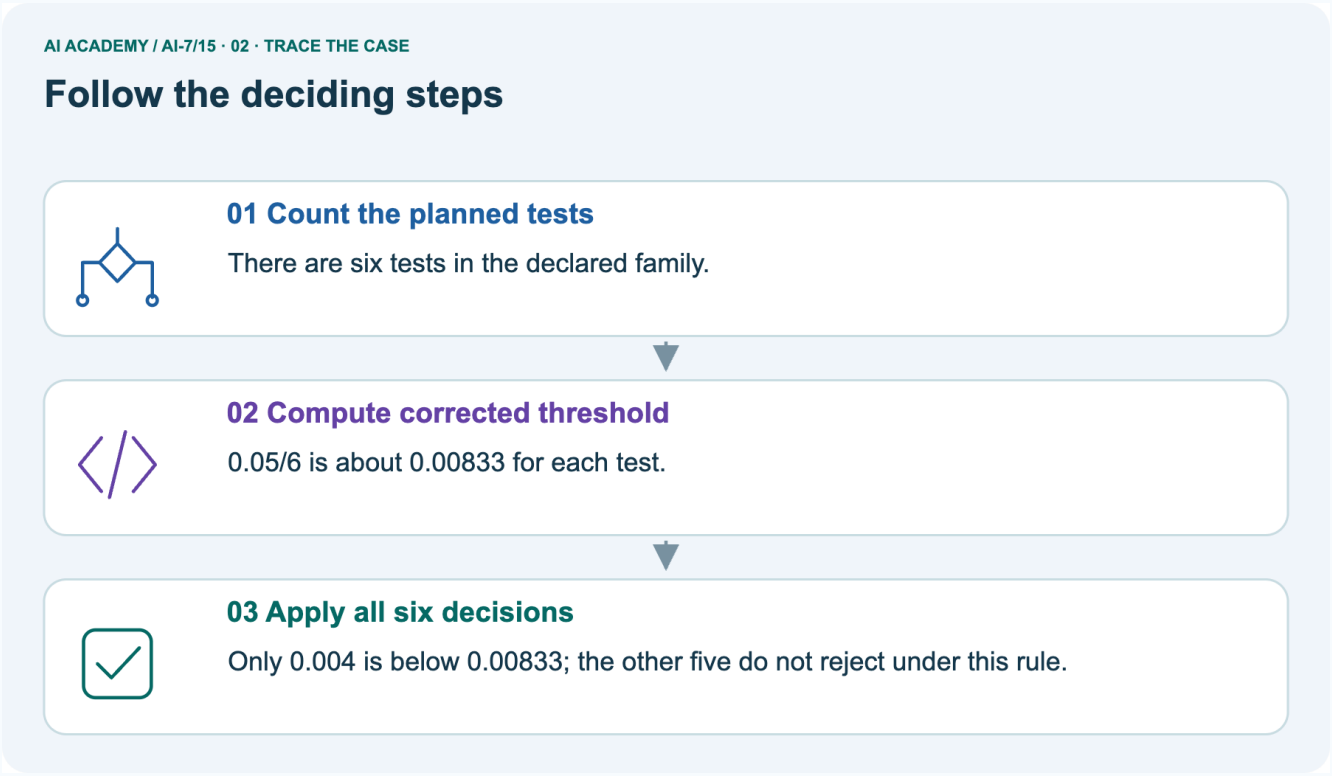
This is a comparison between declared scenarios, not permission to revise the family after seeing favorable results.



Reject a supplied valid test only when p ≤ its family cutoff, including equality.
Ten-test case: four p-values are missing; their decisions remain unknown.
These are decision boundaries, not effect sizes or measured false-positive rates.

Separate picture test	Supplied p	Six-test decision	Ten-test decision
P1	0.004	Reject	Reject
P2	0.012	Do not reject	Do not reject
P3	0.020	Do not reject	Do not reject
P4	0.080	Do not reject	Do not reject
P5	0.400	Do not reject	Do not reject
P6	0.900	Do not reject	Do not reject
Four additional tests	Not supplied	Outside original six-test family	Unknown; remain counted in m = 10

Picture trace and checked change



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Count the planned tests	There are six tests in the declared family.
2. Compute corrected threshold	0.05/6 is about 0.00833 for each test.
3. Apply all six decisions	Only 0.004 is below 0.00833; the other five do not reject under this rule.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Searching many comparisons provides more chances for false positives. Bonferroni divides the error allowance across the declared family; valid individual p-values are required.

A rejected null is not automatically a large or useful improvement.

No effect size or real data is supplied. Bonferroni can be conservative and cannot repair invalid p-values or undisclosed outcome-driven test selection.

Change: The declared family has ten tests instead of six; keep the listed values unchanged.

Prediction: The cutoff becomes 0.005, and only 0.004 among the listed values still rejects.

Reason: The error allowance is divided among ten tests; all ten results must be reported, including the four not supplied here.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Multiple comparisons	Testing several hypotheses in one declared family
Family error	The chance of at least one false rejection under relevant nulls
Bonferroni rule	Dividing a family threshold by the number of tests

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
def bonferroni(p_values, alpha=0.05):
    if not p_values or not 0 < alpha < 1 or any((not 0 <= p <= 1 for p in p_values)):
        raise ValueError('test settings')
    threshold = alpha / len(p_values)
    return (threshold, [p <= threshold for p in p_values])

print(bonferroni([.009, .012, .3, .4, .8]))
```

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?

Thinking move	Teacher prompt
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Multiple comparisons

The facts we will use

AI ACADEMY / AI-7/15

Supplied p-values and six-test decisions

	Supplied p	At cutoff 0.00833
Test 1	0.004	Reject
Test 2	0.012	Do not reject
Test 3	0.02	Do not reject
Test 4	0.08	Do not reject
Test 5	0.4	Do not reject
Test 6	0.9	Do not reject

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/15

Multiple comparisons

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Count the planned tests

There are six tests in the declared family.



2. Compute corrected threshold

0.05/6 is about 0.00833 for each test.



3. Apply all six decisions

Only 0.004 is below 0.00833; the other five do not reject under this rule.

CHANGE ONE THING

The declared family has ten tests instead of six; keep the listed values unchanged.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Five valid tests form one declared family with alpha 0.05. Their p-values are .009, .012, .3, .4, .8. Use Bonferroni: compare each p-value with 0.05/5, including equality. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Threshold .01; only .009 rejected. An unadjusted .05 rule would also reject .012 and change family control.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Multiple comparisons**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Adjustment cannot repair invalid tests, post-hoc outcome selection or unreported analyses.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Calculate the six-test cutoff and identify which supplied p-values reject. Why must the team count every planned comparison rather than report only its smallest p-value?

The cutoff is $0.05/6 \approx 0.00833$. Only $p=0.004$ rejects; the other five exceed the cutoff. Reporting just that result conceals the search and its familywise error control.

Changed-case reasoning: The cutoff becomes 0.005, and only 0.004 among the listed values still rejects. The error allowance is divided among ten tests; all ten results must be reported, including the four not supplied here.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main cutoff 0.01 rejects only 0.009. Picture cutoff 1/120 rejects only 0.004; changed ten-test cutoff 0.005 leaves four unsupplied decisions unknown. Independent four-test cutoff 0.0125 rejects only 0.01, with the remaining 0.02, 0.2 and 0.9 not rejecting. Include equality and retain every declared comparison. Require valid individual tests, without inventing independence, effect sizes or actual observations.

Diagnose the current misconceptions

The worked result establishes every future case: Adjustment cannot repair invalid tests, post-hoc outcome selection or unreported analyses.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Four valid tests form one declared family with alpha 0.05. Their p-values are .01, .02, .2, .9. Use Bonferroni: divide family alpha by four and reject when p is at most that threshold. Compute the threshold and all four decisions.

Teacher explanation: Threshold .0125; reject .01 only. Interpret using the valid-test assumptions, not probability of hypothesis truth.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Testing many hypotheses creates more chances for false discoveries. A family rule must be specified before selecting attractive p-values. For this exercise use Bonferroni: family alpha .05 divided by number of tests m; reject each test when its valid p-value is at most .05/m. This controls the **family error** by a bound without requiring independent tests.

W2 Trace the worked case

Five valid tests form one declared family with alpha 0.05. Their p-values are .009, .012, .3, .4, .8. Use Bonferroni: compare each p-value with 0.05/5, including equality. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Threshold .01; only .009 rejected. An unadjusted .05 rule would also reject .012 and change family control.

W3 Audit the limit

A report omits this qualification: Adjustment cannot repair invalid tests, post-hoc outcome selection or unreported analyses. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Adjustment cannot repair invalid tests, post-hoc outcome selection or unreported analyses.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Adjustment cannot repair invalid tests, post-hoc outcome selection or unreported analyses.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: Threshold .01; only .009 rejected. An unadjusted .05 rule would also reject .012 and change family control. Qualification: Adjustment cannot repair invalid tests, post-hoc outcome selection or unreported analyses.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Four valid tests form one declared family with alpha 0.05. Their p-values are .01, .02, .2, .9. Use Bonferroni: divide family alpha by four and reject when p is at most that threshold. Compute the threshold and all four decisions.

Teacher explanation: Threshold .0125; reject .01 only. Interpret using the valid-test assumptions, not probability of hypothesis truth.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Threshold .0125; reject .01 only. Interpret using the valid-test assumptions, not probability of hypothesis truth. Require intermediate reasoning and the stated scope. Fresh retry: Ten tests have minimum p .006.

Expected reasoning: Threshold .005; no rejection if all other values are larger. Do not hide the other nine tests.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

Add a labelled **multiple comparisons evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **multiple comparisons**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Ablation studies. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: ai-7/16, ai-7/18, ai-7/21. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of **multiple comparisons**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

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>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: Five valid tests form one declared family with alpha 0.05. Their p-values are .009, .012, .3, .4, .8. Use Bonferroni: compare each p-value with 0.05/5, including equality. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Threshold .01; only .009 rejected. An unadjusted .05 rule would also reject .012 and change family control.

Check every learner before discussion: Use the worked case for **Multiple comparisons**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Adjustment cannot repair invalid tests, post-hoc outcome selection or unreported analyses.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The declared family has ten tests instead of six; keep the listed values unchanged. Require a **prediction** and a reason before comparing.

Expected result: The cutoff becomes 0.005, and only 0.004 among the listed values still rejects. Reason: The error allowance is divided among ten tests; all ten results must be reported, including the four not supplied here.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

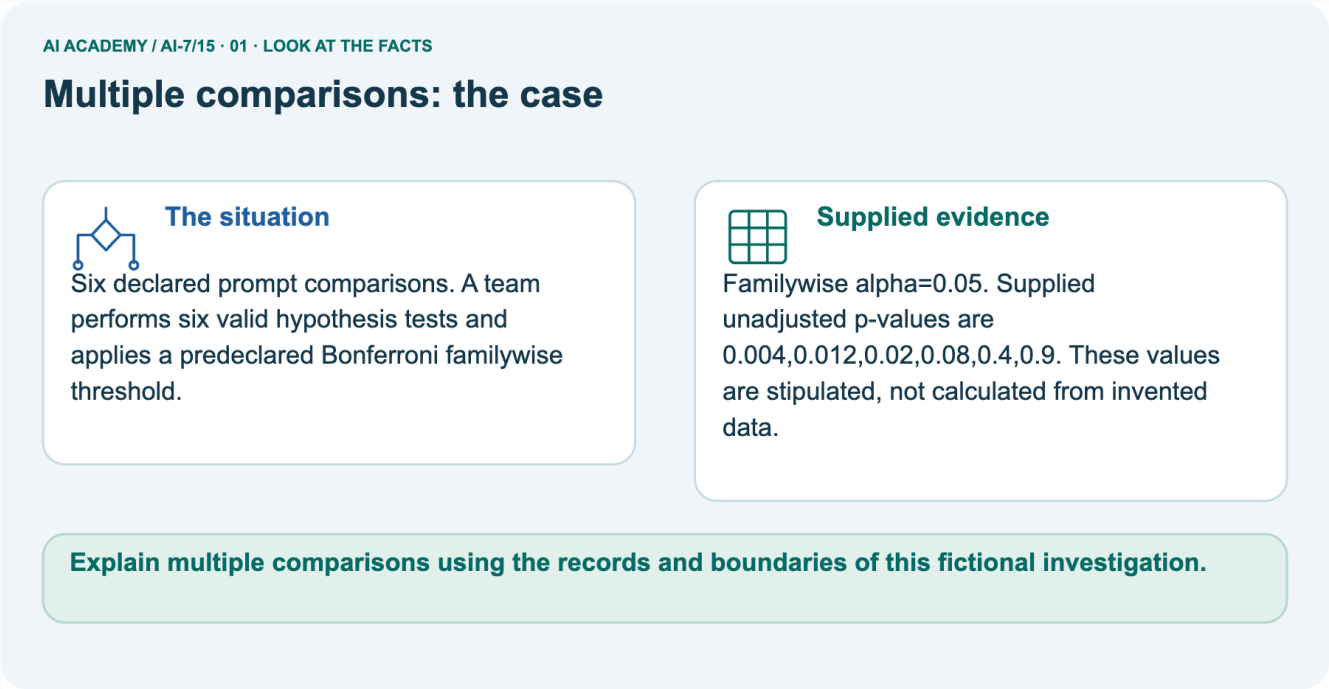
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Multiple comparisons: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

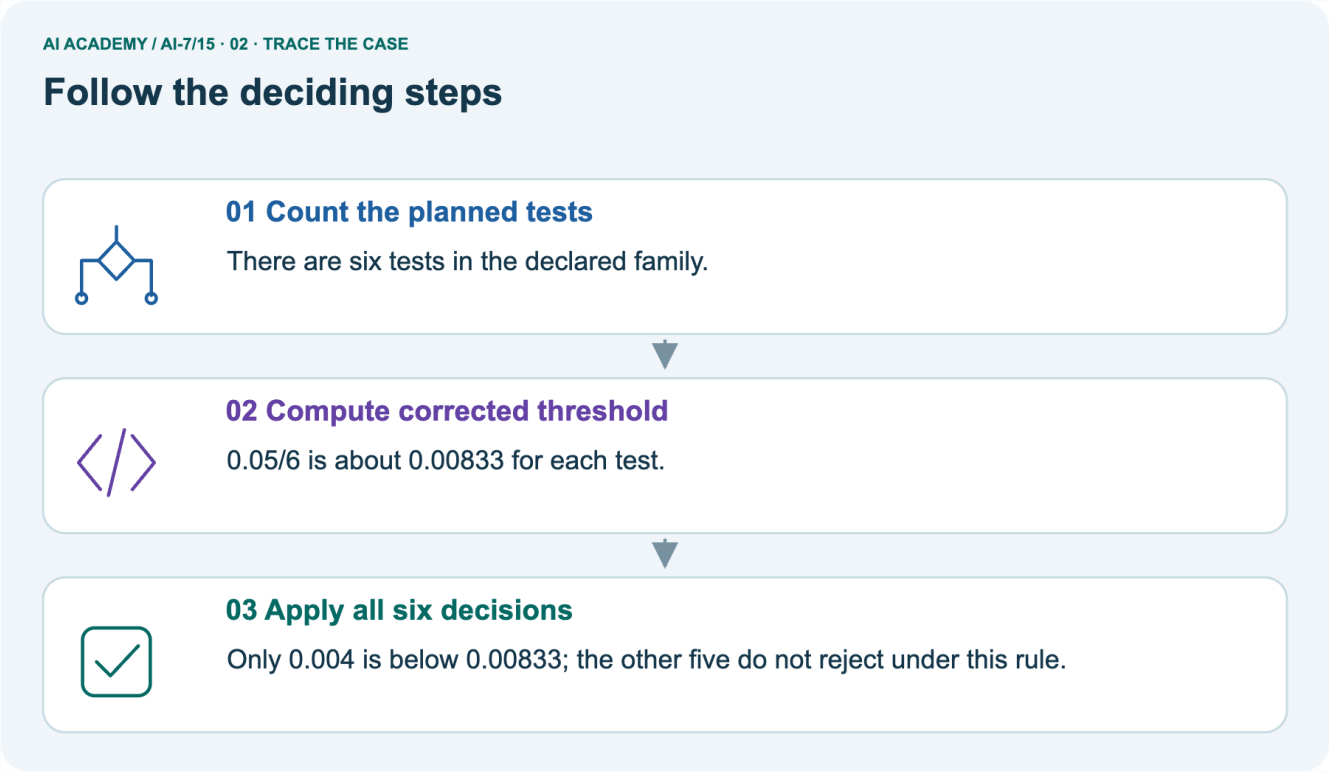
Six declared prompt comparisons. A team performs six valid hypothesis tests and applies a predeclared Bonferroni familywise threshold. Supplied facts: Familywise alpha=0.05.

Supplied unadjusted p-values are 0.004,0.012,0.02,0.08,0.4,0.9. These values are stipulated, not calculated from invented data.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



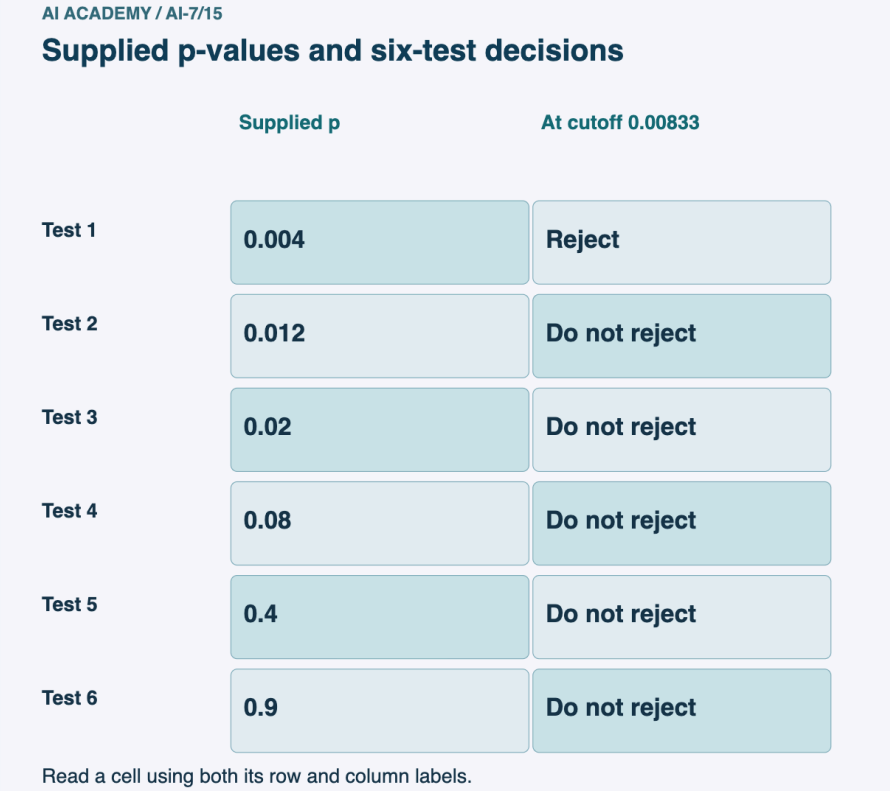
Follow the labelled picture with the matching explanation. Open the vector diagram

Count the planned tests: There are six tests in the declared family. → Compute corrected threshold: 0.05/6 is about 0.00833 for each test. → Apply all six decisions: Only 0.004 is below 0.00833; the other five do not reject under this rule.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Supplied p-values and six-test decisions



Follow the labelled picture with the matching explanation. Open the vector diagram

Supplied p-values and six-test decisions. Test 1: Supplied p = 0.004, At cutoff 0.00833 = Reject. Test 2: Supplied p = 0.012, At cutoff 0.00833 = Do not reject. Test 3: Supplied p = 0.02, At cutoff 0.00833 = Do not reject.

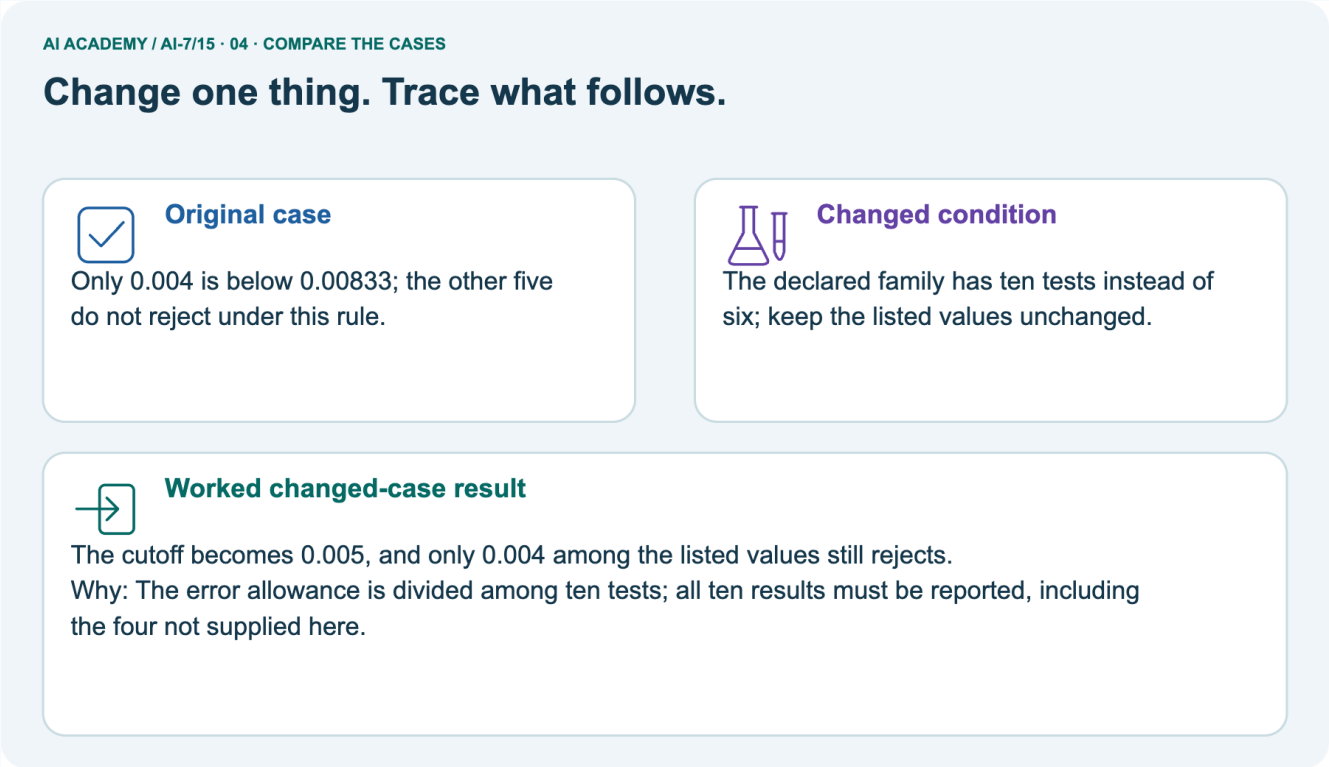
Test 4: Supplied p = 0.08, At cutoff 0.00833 = Do not reject. Test 5: Supplied p = 0.4, At cutoff 0.00833 = Do not reject. Test 6: Supplied p = 0.9, At cutoff 0.00833 = Do not reject.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Only 0.004 is below 0.00833; the other five do not reject under this rule. Changed condition: The declared family has ten tests instead of six; keep the listed values unchanged.

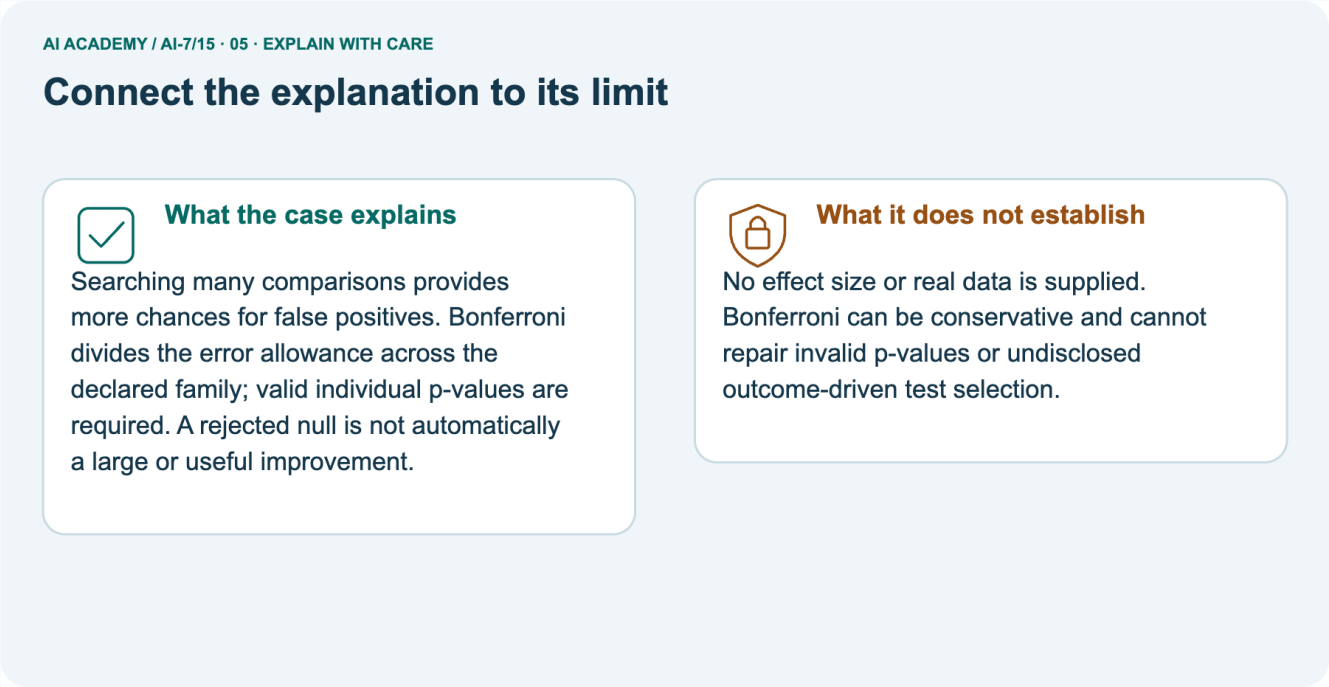
Expected result: The cutoff becomes 0.005, and only 0.004 among the listed values still rejects. Reason: The error allowance is divided among ten tests; all ten results must be reported, including the four not supplied here.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The error allowance is divided among ten tests; all ten results must be reported, including the four not supplied here.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Searching many comparisons provides more chances for false positives. Bonferroni divides the error allowance across the declared family; valid individual p-values are required.

A rejected null is not automatically a large or useful improvement. Limit: No effect size or real data is supplied.

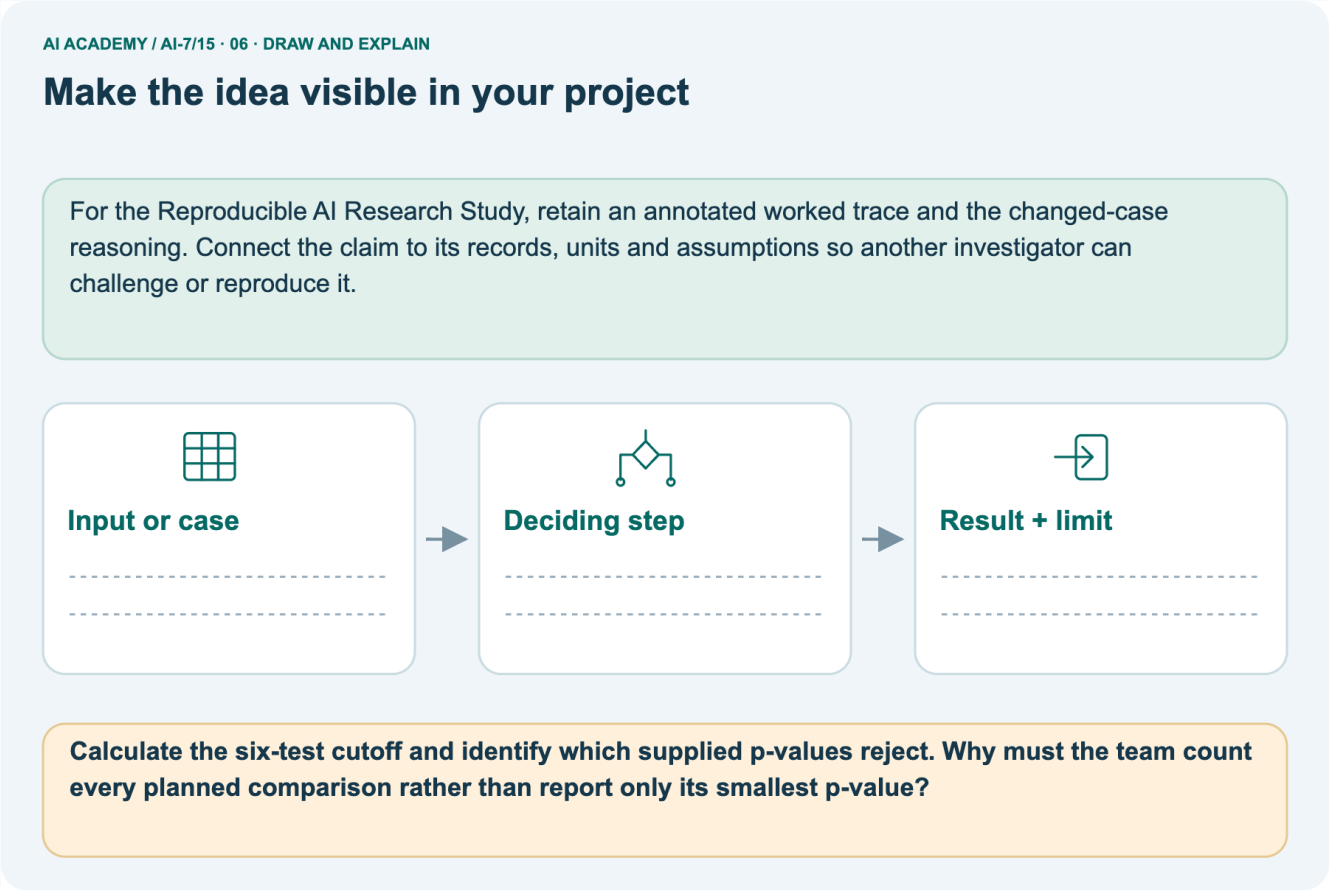
Bonferroni can be conservative and cannot repair invalid p-values or undisclosed outcome-driven test selection.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: No effect size or real data is supplied. Bonferroni can be conservative and cannot repair invalid p-values or undisclosed outcome-driven test selection.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the six-test cutoff and identify which supplied p-values reject.

Why must the team count every planned comparison rather than report only its smallest p-value?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The cutoff is 0.05/6≈0.00833. Only p=0.004 rejects; the other five exceed the cutoff. Reporting just that result conceals the search and its familywise error control.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Multiple comparisons: the case

02. Follow the deciding steps

03. Supplied p-values and six-test decisions

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.